

1. Create Child Table (emp)

```
CREATE TABLE emp(  
    empno INTEGER PRIMARY KEY,  
    empname VARCHAR(20),  
    deptno INTEGER,  
    empsal INTEGER  
);  
DESC student;
```

```
mysql> DESC emp;
```

Field	Type	Null	Key	Default	Extra
empno	int	NO	PRI	NULL	
empname	varchar(20)	YES		NULL	
deptno	int	YES		NULL	
empsal	int	YES		NULL	

4 rows in set (0.02 sec)

2. Insert Data into emp Table

```
INSERT INTO emp VALUES(1,'Ram',5,20000);  
INSERT INTO emp VALUES(2,'Shyam',6,30000);  
INSERT INTO emp VALUES(3,'Ganesh',7,40000);  
SELECT * FROM emp;
```

```
mysql> SELECT * FROM emp;
```

empno	empname	deptno	empsal
1	Ram	5	20000
2	Shyam	6	30000
3	Ganesh	7	40000

3 rows in set (0.01 sec)

3. Create Parent Table (dept)

```
CREATE TABLE dept(  
    deptno INTEGER PRIMARY KEY,  
    dname VARCHAR(10),  
    loc VARCHAR(5)  
);
```

Insert Data into dept Table

```
INSERT INTO dept VALUES(5,'IT','hyd');  
INSERT INTO dept VALUES(6,'sales','bglr');  
INSERT INTO dept VALUES(7,'Manager','mumb');
```

SELECT * FROM dept;

```
mysql> SELECT * FROM dept;
+-----+-----+-----+
| deptno | dname  | loc   |
+-----+-----+-----+
|      5 | IT     | hyd   |
|      6 | sales  | bglr  |
|      7 | Manager | mumb  |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

4. Add Foreign Key Constraint

```
ALTER TABLE emp
ADD CONSTRAINT fk
FOREIGN KEY(deptno) REFERENCES dept(deptno);
```

5. Test Constraint (Error Case)

INSERT INTO emp VALUES(8,'Ram',8,50000);

```
mysql> INSERT INTO emp VALUES(8,'Ram',8,50000);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`dbms5`.`emp`, CONSTRAINT `fk` FOREIGN KEY (`deptno`) REFERENCES `dept` (`deptno`))
mysql> |
```

6. Drop Foreign Key Constraint

ALTER TABLE emp DROP CONSTRAINT fk;

```
mysql> ALTER TABLE emp DROP CONSTRAINT fk;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

7. Insert Data After Dropping Constraint

INSERT INTO emp VALUES(8,'Ram',8,50000);

```
mysql> INSERT INTO emp VALUES(8,'Ram',8,50000);
Query OK, 1 row affected (0.02 sec)
```

8. Final Display

SELECT * FROM emp;

SELECT * FROM dept;

```
mysql> SELECT * FROM emp;
+-----+-----+-----+-----+
| empno | empname | deptno | empsal |
+-----+-----+-----+-----+
|      1 | Ram     |      5 | 20000  |
|      2 | Shyam   |      6 | 30000  |
|      3 | Ganesh  |      7 | 40000  |
|      8 | Ram     |      8 | 50000  |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM dept;
+-----+-----+-----+
| deptno | dname   | loc   |
+-----+-----+-----+
|      5 | IT      | hyd   |
|      6 | sales   | bglr  |
|      7 | Manager | mumb  |
+-----+-----+-----+
3 rows in set (0.00 sec)
```